Unit no. ° 9

PROPORTIONS AND CHANGES

21-29 January 2027

I Proportion and proportion of a proportion

Definitions

In a non-empty finite set E, the proportion of a subset F is $p = \text{Card}(F)/\text{Card}(E)$; $0 \leq p \leq 1$. Expressed as a percentage, it is $100\,p$ %. If $G \subseteq F \subseteq E$, $F \neq \varnothing$, and if p_1 is the proportion of F in E, p_2 that of G in F, the proportion of G in E is $p_1 p_2$.

Example 9.1

A lycée has 1500 students, 652 of whom are half-boarders. Calculate their proportion. In one class, 60 % of the students do sport; among them, 40 % are boys. What proportion of the class do the boys who do sport represent?

Solution:

$652/1500 \approx 0{,}4347$, i.e. about 43 ,47%. For the class, 0 $,60 \times 0{,}40 = 0{,}24$, i.e. 24%. The second percentage refers to the students who do sport, not the whole class.

Example 9.2

In a fictional school of 800 pupils, 35% do a club sport; among them, 45% do a team sport. Calculate the two numbers and the final proportion.

Solution:

$800 \times 0{,}35$ = 280 pupils in a club. 280 $\times$ 0,45 = 126 in team sports. The final proportion is $126/800 = 0{,}1575 = 15{,}75\,\%$.

Watch out. The proportion and the rate of change do not answer the same question: a proportion compares a part to a set, a rate compares a variation to an initial value.

II Absolute change and rate of change

Definition

For an initial value $V_I > 0$ and a final value $V_F > 0$, the absolute change is $V_F - V_I$; the rate of change is

$$t = \frac{V_F - V_I}{V_I}.$$

The rate has no unit. It is positive for an increase, negative for a decrease. The reference is always the initial value.

Example 9.3

A cheese is sold at 2 ,13 euros instead of 3 ,55 euros. Calculate the absolute change and the rate.

Solution:

Absolute change: 2 $,13 - 3,55 = -1,42$ euros. Rate: $-1,42/3,55 = -0,40$, i.e. a decrease of 40%.

Multiplier

$CM = V_F/V_I = 1 + t$, therefore $V_F = V_I(1 + t)$. To recover V_I, divide V_F by $1 +$ t (not zero).

Example 9.4

A hypothetical item costing 50000 Ar goes up by 12 %, then we consider separately a 12 % discount on the initial price. Give the final prices. A price after a 20% increase is 72000 Ar; find its initial.

Solution:

Increase: 50000 $\times$ 1,12 = 56000 Ar. Discount: 50000 $\times$ 0,88 = 44000 Ar. Last question: $V_I = 72000/1,20$ = 60000 Ar.

Fluency · 5 to 10 minutes

Expand ($x - 3)^2$; solve 1 $/x$ = 5; give the mean of 8; 10; 12.

Answers: $x^2 - 6x + 9$; $x = 1/5$; 10.

III Successive percentage changes

Property

Successive changes at rates t_1, t_2 have overall multiplier

$$CM_G = (1+t_1)(1+t_2), \qquad t_G = (1+t_1)(1+t_2) - 1.$$

We multiply the multipliers; we do not simply add the rates.

Example 9.5

A manufacturer sold 5600 bicycles in 2022, i.e. 25 % more than in 2021. Its sales fall by 15 % in 2023. Determine the sales in 2021 and 2023, then the overall rate between those two years.

Solution:

$2021 : 5600/1{,}25$ = 4480 bicycles. 2023: $5600 \times 0{,}85$ = 4760 bicycles. Overall multiplier $1{,}25 \times 0{,}85 = 1{,}0625$, so an overall increase of 6 ,25%. Check: (4760 $- 4480)/4480 = 0{,}0625$.

Example 9.6

A price rises by 10 % then falls by 10 %. Does it return to the original price?

Solution:

$CM_G = 1{,}10 \times 0{,}90 = 0{,}99$. The final price is 99% of the initial price: overall decrease of 1%. The two rates relate to different reference values.

Example 9.7

A rate rises from 20 % to 25 %. Explain the difference between the increase in percentage points and relative change.

Solution:

The difference is 5 percentage points. The relative change is (0 $,25 - 0{,}20)/0{,}20 = 0{,}25$, i.e. a increase of 25%.

IV Reverse percentage change

Property

To return to the initial value after a change at rate $t > -1$, use the multiplier $1/(1+t)$. The reverse rate is

$$t_{\text{reverse}} = \frac{1}{1+t} - 1 = \frac{-t}{1+t}.$$

Example 9.8

What rate offsets a 25% increase? What rate offsets a 20% decrease?

Solution:

After an increase of 25%, the reverse multiplier is $1/1{,}25 = 0{,}8$: a decrease of 20%. After a decrease of 20%, it is $1/0{,}8 = 1{,}25$: an increase of 25%.

Example 9.9

A price is reduced by 30 %, then rises by 10 %. What additional percentage increase restores the original price?

Solution:

The current multiplier is $0{,}70 \times 1{,}10 = 0{,}77$. The multiplier sought is $1/0{,}77 = 100/77$. The rate is $23/77 \approx 0{,}2987$, i.e. about 29,87%. Check $0{,}77 \times (100/77) = 1$.

Guidance for the teacher

Have students state the base of the percentage at each step. The formula for the reverse rate is justified by a product equal to 1; memorisation alone is fragile.